GRAND ROUNDS CASE · FOLLOW-UP PROGRAM

Followup 26-12

Viral exanthem with respiratory distress

Patient	37-year-old female. Obesity, former smoker, no other significant history. **Seen in an emergency department three days earlier** for cough, fever, rash, vomiting and body aches of three days' duration. White count 17 and a clear chest radiograph, treated as a presumed pneumonia on the strength of wheeze, and discharged on azithromycin, a three-day course of prednisone, and an albuterol inhaler.
Outcome	Survived. **Not anaphylaxis.** Respiratory film array detected **rhinovirus or enterovirus**. She was admitted with sepsis of viral origin, community-acquired pneumonia not excluded, and the rash was read as a **viral exanthem** on the grounds that it began before the antibiotic was ever started.
Disposition	Treated with methylprednisolone, nebulized ipratropium and albuterol, fluids, ceftriaxone and doxycycline. Admitted and discharged the same day.

The call

Neighbors called after a text message from inside the residence that said "help." Two units responded, LALS first and a paramedic unit about seven minutes behind. Times run from LALS patient contact.

She was on the living room floor in obvious respiratory distress, coughing and vomiting into a coffee can, with diffuse hives across her body. She could speak only in one to two word sentences. Saturation 89 to 90% on room air. Lung sounds markedly diminished bilaterally, faint to nearly silent.

- **+0:** Airway, breathing, circulation. Oxygen 6 L by nasal cannula.
- **+3:** Epinephrine 0.3 mg intramuscular, left deltoid.
- **+4:** Blood pressure unobtainable. Pulse 110, respirations 30, SpO2 90%.
- **+7:** Paramedic patient contact.
- **+11:** Albuterol 2.5 mg nebulized. An intravenous attempt with a 22 gauge in the hand was unsuccessful.
- **+20:** 147/88, MAP 108, pulse 102, respirations 20, SpO2 94%.
- **+24:** IV access.
- **+34:** 139/98, MAP 112, pulse 100, SpO2 98%.

The paramedic's history changed the picture. The rash had been present for several days. No exposure to any known allergen or new substance. A recent pneumonia, which the patient herself blamed for the rash. On examination the rash was red and patchy on all four extremities and **not consistent with hives**, with crackles in the left lower lobe and clear fields elsewhere.

She walked from the stretcher to the hospital bed on arrival.

Primary impression

Anaphylaxis.

The hospital did not find one. She arrived tachycardic and tachypneic, saturating 92% on room air, coarse with bilateral expiratory wheeze and a diffuse rash. The septic workup found **rhinovirus or enterovirus** on respiratory film array, a white count risen from 17.6 to 20.5, a clear chest film, and a procalcitonin of 0.10, which the laboratory's own comment places where bacterial infection is possible but unlikely. She was admitted with sepsis of viral origin and discharged the same day.

The discharge summary is specific about the rash: it began **before** the azithromycin was started. That timing rules out the drug reaction and points at the virus she had been carrying for the better part of a week.

Education points

What anaphylaxis requires

Anaphylaxis is a clinical diagnosis with a shape, and the shape is worth knowing precisely, because the presentations that imitate it are common and the one that is real is uncommon.

It is acute. The whole entity is a sudden, systemic release of histamine and other mediators from mast cells and basophils, and it happens over minutes to a couple of hours. That is not a stylistic feature, it is the mechanism. Mediator release is fast and short-lived, which is why anaphylaxis presents abruptly and why untreated cases either kill or resolve rather than sitting still for days.

It follows an exposure. Food, drug, sting, latex, contrast. Sometimes the trigger is never identified, and idiopathic anaphylaxis is real, but an exposure history is the single most useful piece of information available and its absence is evidence rather than a gap in the story.

It involves more than one organ system, typically skin or mucosa plus respiratory compromise or hypotension, or hypotension alone after a known allergen.

Set this patient against that. The skin findings were **five or six days old** and had already been examined and documented by a physician three days earlier. No exposure was found by anybody at any point. She had a known viral illness, a known reactive airway, and had been on albuterol and prednisone for two days without improving.

The first criterion fails on the rash alone. A rash that has been in the same place since the weekend is not the cutaneous limb of an acute reaction, and urticaria that old will not become anaphylaxis. Whatever was making her breathless that morning, it was not a mast cell event that started three minutes before anybody arrived.

Reading the rash

Three experienced examiners looked at this rash and gave three answers. A physician three days earlier documented annular raised blanchable patches across her entire body, **consistent with urticaria**. The paramedic on this call documented red patchy lesions **not consistent with hives**. The admitting physician concluded **viral exanthem**.

The question that separates them is not what it looks like. It is **how long an individual lesion lasts.**

A true urticarial wheal is transient. Individual lesions come up, migrate, and resolve **within twenty-four hours leaving no mark**, while new ones appear elsewhere. That is the defining feature, and it is the one thing morphology cannot tell you. A viral or drug exanthem is fixed: the same lesions sit in the same places for days and fade gradually.

It is a ten-second question on scene. **Are these the same spots that were there yesterday, or new ones?**

Hers had been in the same distribution for several days, which is not urticaria however raised or blanchable it looked on any given day.

> **Further reading on urticarial rashes.** The differential between urticaria, viral exanthem, drug exanthem and the dangerous mimics is worth twenty minutes. This open-access review covers it clearly: https://www.frontiersin.org/journals/allergy/articles/10.3389/falgy.2022.808543/full

What she actually had

A viral respiratory infection in a reactive airway, two days into treatment that was not working.

The virus was rhinovirus or enterovirus, a family that causes most common colds and, in some people, a great deal more than that. In an airway that responds to inflammation with bronchospasm, a viral infection is one of the most reliable triggers there is. She had been prescribed albuterol and prednisone three days earlier for exactly this, which tells you somebody had already recognised the airway component.

The rash was the same virus. A viral exanthem is a rash produced by the infection itself, through immune response and direct viral effect on the skin, and it is typically widespread, erythematous, and not itchy. Hers was documented as widespread, erythematous and **nonpruritic** at the hospital. That last detail is worth holding onto, because urticaria itches and this did not.

The treatment that worked was bronchodilation and steroid. Nebulized ipratropium with albuterol, methylprednisolone 125 mg, and fluids, with work of breathing documented as significantly improved within about ninety minutes. The white count and the low procalcitonin got her antibiotics as a hedge, but the thing that made her better was opening the airway and suppressing the inflammation driving it closed.

Prehospital considerations

Ask how old the individual spots are. Not how long the rash has been there, which she was asked and answered. How long **each** lesion lasts. That question sorts hives from everything that imitates them, and it takes ten seconds.

Ask whether it itches. Urticaria itches, reliably and often intensely. A widespread rash that does not itch is pointing somewhere else.

Find the exposure, or notice that there isn't one. No exposure was identified on this call, by anybody, at any point. A missing trigger does not by itself exclude anaphylaxis, but it is a substantial piece of evidence and it deserves to be weighed rather than passed over.

A recent, unresolved illness is context, not background. She had been seen three days earlier, given a diagnosis and three medications, and was not better. That fact was available on scene and it reframes everything that follows it. A patient who is failing outpatient treatment for a known problem is usually failing at that problem rather than developing a new one.

Document the pre-treatment examination and hand it over. The first crew's record of a nearly silent chest and an unobtainable blood pressure is the only account anywhere of how she looked before anything was given. Whatever else is true of this call, that record exists because somebody wrote it down while it was happening.

The interventions

PREHOSPITAL INTERVENTIONS

Oxygen at 6 L by nasal cannula, for a saturation of 89 to 90% on room air.

Epinephrine 0.3 mg intramuscular, left deltoid. The site and route are documented, which matters because absorption differs between deltoid and thigh and because a recorded site tells the hospital where to look if a local reaction develops.

Albuterol 2.5 mg nebulized. A beta-2 agonist relaxing bronchial smooth muscle directly. It is the treatment for the bronchospasm she actually had, and the hospital gave it again with ipratropium within the hour.

Intravenous access, attempted unsuccessfully with a 22 gauge by the first crew and established by the paramedic.

Transport to the patient's requested facility, which was appropriate and cost nothing.

IN-HOSPITAL INTERVENTIONS

Respiratory film array. The test that found the answer, screening a panel of respiratory viruses and atypical bacteria by nucleic acid amplification. Three days earlier a narrower panel covering COVID, influenza and RSV had been negative, which is why the virus went unidentified on the first visit. **A negative COVID and flu swab is not a negative viral workup.**

Methylprednisolone 125 mg IV, with nebulized ipratropium and albuterol and fluids.

Ceftriaxone and doxycycline, covering community-acquired pneumonia including atypical organisms, given despite a low procalcitonin and a clear chest film because a rising white count in an unwell patient is not left uncovered while cultures are pending.

The labs, and what they mean

Reference ranges vary between institutions and analyzers. These are the ranges printed on the source reports.

Lab	Three days before	Day of call	Reference range
Rhinovirus or enterovirus PCR	Not tested	**Detected**	Not detected
White blood cell (x10^9/L)	17.6	20.5	4.0 to 10.8
Neutrophils, absolute (x10^9/L)	14.30	15.11	1.80 to 7.80
Lymphocytes (%)	11.9	18.3	20.0 to 50.0
Procalcitonin (ng/mL)	Not tested	0.10	0.02 to 0.09
Potassium (mmol/L)	3.3	3.2	3.4 to 5.0
Bicarbonate (mmol/L)	19	not reported	22 to 29

The white count rose from 17.6 to 20.5 across three days, and some of that rise is the prednisone. Read alone it looks like an infection worsening. She had been taking 40 mg of prednisone daily since the day before this call, and corticosteroids raise the white count and lower the lymphocyte count on their own, pushing neutrophils out

of the marrow and out of the vessel walls while pulling lymphocytes out of circulation. The record does not separate drug effect from disease and neither can anybody else.

Procalcitonin rises in bacterial infection and stays low in viral illness. Hers was 0.10, and the laboratory comment places anything between 0.05 and 0.25 where bacterial infection is possible but unlikely. Against a clear chest film and a positive viral panel, that is a consistent picture and it is why the discharge summary names a virus.

Two values worth reading on your own. Her **potassium ran 3.3 then 3.2**, mildly low both times. Beta-2 agonists drive potassium into cells, so a patient already on scheduled albuterol who then receives epinephrine and a nebulizer is pushed further down. Neither value needed treatment. Her **bicarbonate of 19** on the first visit, with a normal lactate, fits days of fever, vomiting and poor intake rather than anything acute.

Learning points

1. **Anaphylaxis is acute, and hers was not.** The mechanism is a sudden release of mast cell mediators over minutes to a couple of hours. A rash that has been in the same place for five days is not part of that event, and urticaria that old will not become anaphylaxis.
2. **Ask how long each spot lasts, not how long the rash has been there.** A wheal resolves inside twenty-four hours and leaves nothing behind; a viral or drug exanthem sits in place for days. Morphology gave three experienced examiners three different answers here. Duration would have given one.
3. **Ask whether it itches.** Urticaria itches. Hers was documented at the hospital as nonpruritic, which is a viral exanthem and not a histamine rash.
4. **No exposure is evidence.** Nobody on this call found a trigger at any point, and the answer turned out to be a virus she had been carrying for the better part of a week. A missing exposure does not close the door on anaphylaxis, but it should move the differential.
5. **A patient failing outpatient treatment is usually failing at the original problem.** She had been seen three days earlier, given three medications for a reactive airway and a presumed pneumonia, and had not improved. That history was available on scene and it explains everything that happened afterward.

Prepared from records obtained through the follow-up request process and de-identified for education. To request follow-up on a call you ran, fill out the request form at https://bcmca.github.io/followup/